

Refusal Is Not Free: Measuring Hallucination and Over-Refusal in a Retrieval-Augmented Waste-Segregation Chatbot for SDG 11 and SDG 12

Mark Emmanuel P. Sagarino
College of Computer Studies
Course ITC-C508
Mandaluyong City, Philippines
 markemmanuel.sagarino@my.jru.edu

Abstract—This paper documents the construction and adversarial evaluation of a Retrieval-Augmented Generation (RAG) chatbot for Philippine household solid-waste segregation, a domain aligned with SDG 11 and SDG 12. A corpus of three curated documents was split into 38 chunks, embedded with all-MiniLM-L6-v2 and indexed in ChromaDB, then queried through openai/gpt-oss-20b under a strict grounding contract. Across four in-domain and one out-of-domain query the system produced zero factual hallucinations. Two results complicate the usual conclusion. First, the grounding contract caused a false refusal on an in-domain question whose answer was present in the corpus but absent from the top-3 retrieval, showing that refusal has a measurable recall cost. Second, removing the fallback instruction and raising temperature to 1.0 did not induce hallucination; it produced four different refusal wordings in four generations. The instruction’s measured benefit was therefore output determinism, not hallucination suppression.

Keywords—*retrieval-augmented generation; hallucination; over-refusal; vector embeddings; prompt grounding; SDG 11; SDG 12; solid-waste management.*

I. SYSTEM ARCHITECTURE

The chatbot answers questions about household solid-waste segregation, recycling and the Philippine Ecological Solid Waste Management Act [1]. The domain was chosen because it sits at the intersection of SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production) [2], and because it is a domain where a wrong answer has a real cost: a citizen told the wrong container colour or the wrong penalty figure acts on that error. This makes hallucination [5] the primary risk to engineer against.

The system is a two-phase Retrieval-Augmented Generation pipeline [3], shown in Fig. 1: an offline ingestion phase that runs once and turns the corpus into a searchable vector index, and an online phase that runs per query and is what the user experiences.

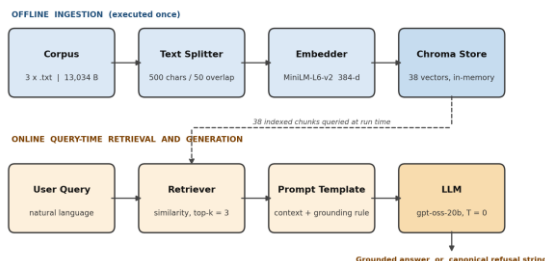


Fig. 1. Two-phase RAG architecture. The ingestion path (top) is executed once; the query path (bottom) executes per question against the in-memory 38-vector index.

Ingestion runs in four steps. A DirectoryLoader reads three UTF-8 documents totalling 13,034 bytes: a plain-language summary of Republic Act 9003, a city segregation guide covering the four container categories and the collection schedule, and a ten-item recycling FAQ. A RecursiveCharacterTextSplitter divides them into 38 chunks, each embedded by all-MiniLM-L6-v2 [4] into a 384-dimensional vector and written to an in-memory ChromaDB collection.

At query time the question is embedded by the same model and the three nearest chunks by cosine similarity are retrieved. create_stuff_documents_chain concatenates them into a {context} placeholder in a ChatPromptTemplate, and create_retrieval_chain wires retriever and generator into one callable. The prompt carries the grounding contract: answer only from the supplied context, and when it is insufficient emit the exact sentence “I cannot answer based on the provided domain data.” Generation runs on openai/gpt-oss-20b hosted on Groq.

II. IMPLEMENTATION DETAILS

Table I lists every parameter governing system behaviour. Values follow the course reference implementation [6] so that the evaluation isolates the effect of the prompt contract rather than of retrieval tuning.

TABLE I. SYSTEM PARAMETERS AND CONFIGURATION

Parameter	Value	Rationale / Effect
Chunk size	500 characters	Keeps one topic per chunk; smaller values fragment enumerated lists.
Chunk overlap	50 characters	Preserves sentences that straddle a chunk boundary.
Chunks produced	38	From 3 documents, 13,034 bytes total.
Embedding model	all-MiniLM-L6-v2	384-dim; runs locally, no API cost.
Vector store	Chroma (in-memory)	Rebuilt each session; no persistence needed at this scale.
Retriever	similarity, k = 3	Caps prompt length; the binding constraint in Section III-B.
LLM	openai/gpt-oss-20b	Hosted on Groq; see the deviation note below.
Temperature (baseline)	0.0	Greedy decoding for reproducibility.
Temperature (ungrounded)	1.0	Stress-test condition in Section III-C.
Credential handling	getpass at runtime	Key never written to disk or notebook output.

a. All parameters were held fixed across every experiment except temperature and the presence of the fallback instruction, which are the two independent variables.

A. Deviations from the Reference Template

Two substitutions were necessary. The template’s import of langchain.chains fails on LangChain 1.x, which removed the legacy chain builders, so all langchain packages were pinned below version 1.0. The template’s model llama-3.1-8b-instant returns HTTP 404 model not found because Groq deprecated it for the free tier; rather than hard-code a replacement that could itself expire, the notebook queries the Groq /v1/models endpoint at run time and selects the first available model from a ranked preference list, resolving to openai/gpt-oss-20b out of the 14 offered.

B. API Key Protection

The Groq credential is read through `getpass.getpass`, which accepts input without echoing it and holds it only in the process environment for the life of the runtime. The key never appears in the notebook source, in any cell output, or in the saved .ipynb, so the file can be submitted or shared without leaking it.

III. EXPERIMENTAL RESULTS AND HALLUCINATION ANALYSIS

A. In-Domain Accuracy

Four in-domain questions were posed under the baseline configuration. Table II reports the expected answer, the observed behaviour and a verdict. No answer contained a fact absent from the corpus, so hallucinations number zero of four; completeness, only two.

TABLE II. IN-DOMAIN QUERY RESULTS (TEMPERATURE = 0, FALLBACK RULE PRESENT)

Query	Expected Answer	Actual Answer	Verdict
Q1. Four waste categories and their container colours	Biodegradable/green, Recyclable/blue, Residual/black, Special-hazardous/red (doc 02).	Emitted the fallback string "I cannot answer based on the provided domain data."	False refusal. All 3 retrieved chunks came from doc 02, yet none held the category list.
Q2. Penalty for open burning under RA 9003	Fine of PHP 300–1,000 and/or 1–15 days imprisonment (doc 01).	"a fine ranging from P300 to P1,000, or imprisonment of not less than — the exact imprisonment term is not specified in the provided context."	Partially correct. Fine exact; the imprisonment term is in the corpus but was not retrieved.
Q3. What is a Materials Recovery Facility and who must establish one	A sorting and recovery facility; required of every barangay or cluster of barangays (doc 01).	Correct definition; "every barangay, or a cluster of barangays."	Correct and fully grounded.
Q4. Are plastic sachets recyclable, and why	No — bonded multi-layer plastic, classed as residual waste; refuse at purchase (docs 02, 03).	No — bonded thin layers cannot be economically separated; residual waste; refuse at purchase, buy in bulk or refill.	Correct and fully grounded.

b. Hallucinated facts: 0 of 4. Fully correct: 2 of 4. Partially correct: 1. False refusals: 1.

B. Diagnosing the False Refusal

Q1 is the most instructive result, because the system failed on a question it should have answered. The retrieval trace shows all three chunks came from the correct source document, so the embedding model located the right topic — but it returned the composting and collection-schedule passages, not the enumerated category list. The cause is an interaction between two parameters in Table I: the list occupies roughly 1,400 characters, so a 500-character chunk spreads it over three chunks, none individually a strong match for a question phrased around "four categories" and "colour," and with k fixed at 3 the competing passages crowded them out. Q2 fails the same way in milder form: the imprisonment term appears verbatim in 01_RA9003_solid_waste_law.txt, and that passage was in fact retrieved during the out-of-domain test in Section III-C, yet it was not among the top three for the query that actually needed it.

This isolates the cost side of a strict grounding contract. The contract cannot distinguish "the corpus does not contain this" from "retrieval did not surface it," so a retrieval miss becomes a confident refusal — safe, but wrong, and indistinguishable to the user from a genuinely silent corpus. Both failures are retrieval-side, and both are fixable without touching the prompt: raise k to 5, increase chunk size to about 1,000 characters so enumerated lists stay intact, or switch to maximal marginal relevance.

C. Hallucination Stress-Test

The out-of-domain probe asked for the average Earth–Mars distance and the duration of a crewed mission — a fact the corpus cannot contain and the model certainly knows from pre-training. The same question was put to two configurations sharing one retriever, so the retrieved chunks were identical and the prompt was the only difference.

TABLE III. BASELINE VERSUS UNGROUNDED CONFIGURATION ON THE SAME OUT-OF-DOMAIN QUERY

Configuration	Temp.	Fallback rule	Response
[A] Baseline	0.0	Present	"I cannot answer based on the provided domain data." (canonical string)
[B] Ungrounded	1.0	Removed	"I'm sorry, but the information you're asking for isn't included in the context provided."
[B] Repeat 1	1.0	Removed	"I'm sorry, but I can't provide that information."
[B] Repeat 2	1.0	Removed	"I'm sorry, but the provided context does not include any information about the average distance from Earth to Mars..."
[B] Repeat 3	1.0	Removed	"I'm sorry, but I don't have that information."

c. Retrieved chunks were byte-identical across all five generations. Hallucinated facts: 0 of 5. The four ungrounded generations produced four different wordings, no two alike, ranging from an eight-word refusal to a full restatement of the question.

The expected outcome was that configuration B would answer from parametric memory and hallucinate a distance. It did not: the model refused in all four ungrounded generations. The correct reading is not that prompt engineering was unnecessary, but that the retrieved context — visibly about composting and schedules — was itself a strong enough signal for an instruction-tuned model to decline.

What the rule demonstrably changed is the form of the refusal. Configuration A returned one exact, pre-specified sentence; configuration B returned four different wordings in four generations, ranging from an eight-word refusal to a full restatement of the question. For a deployed system this is not cosmetic: a canonical refusal string can be caught by an exact string match and routed to a human operator or a logged unanswered-question queue; unpredictable paraphrases cannot. The measured value of the fallback instruction here is therefore determinism and machine-parseability rather than hallucination suppression, and temperature 0 reinforces it by making decoding greedy and reproducible.

IV. CONCLUSION

The chatbot was evaluated against four in-domain and one out-of-domain query, and produced no hallucinated facts in any configuration tested. Three lessons follow. Prompt restrictions worked, but not as anticipated: the ungrounded model committed no hallucination for the fallback rule to prevent; what the rule delivered was a single deterministic refusal string in place of four unpredictable paraphrases, which is what makes an unanswerable query detectable in production. Retrieval grounding was the real constraint on quality, and failed in both directions: it correctly withheld an answer the corpus lacked, and withheld two it contained, because chunk size and k were tuned for prompt economy rather than recall. Refusal is therefore not free, and reporting only the zero-hallucination figure would hide a false-refusal rate of one in four — a proportion that, on four queries, marks a real failure mode rather than a precise rate. Credential protection used `getpass`, so the API key lives only in runtime memory and never enters the saved notebook. Future work is retrieval-side: raise k , enlarge chunks, add a reranking stage — so the contract refuses only when the corpus is genuinely silent.

REFERENCES

- [1] Republic of the Philippines, Republic Act No. 9003: Ecological Solid Waste Management Act of 2000, Manila, Philippines, 2001.
- [2] United Nations, Transforming Our World: The 2030 Agenda for Sustainable Development, A/RES/70/1, New York, NY, USA, 2015.
- [3] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. NeurIPS, 2020, pp. 9459–9474.
- [4] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992.
- [5] Z. Ji et al., "Survey of hallucination in natural language generation," ACM Comput. Surv., vol. 55, no. 12, pp. 1–38, 2023.
- [6] R. C. Raga Jr., "Retrieval-Augmented Generation," ITC 508 course lecture, 2026.